

James Dominguez

AI ENGINEER

Seattle, WA

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"In algorithms, as in life, persistence usually pays off." — Steven S. Skiena

Summary

Professional AI Engineer, writing mainly Python. Proficient at processing complex data and generating reports using models. Deep understanding of the entire stack with a working knowledge of many modern architectures and technologies. Solid UI skills with the toolset to generate beautiful UIs. Strong written and oral english. Adept at group work; comfortable quickly learning new frameworks and codebases.

Skills

AI/ML	NumPy, PyTorch, Hugging Face, instructor, RAG, knowledge graphs
Languages	Python, Golang, TypeScript, Ruby, Lisp, Racket
Backend	Python, Golang, PostgreSQL, FastAPI, Chroma, Neo4j, LanceDB
Frontend	Vircel v0, shadcd/ui, Tailwind CSS, React etc.
Tools	Vim, Tmux, Awk, Jupyter/Google Colab, LLMs, VS Code
DevOps	Kubernetes/K8s, AWS, Docker, Terraform, Pulumi, Linux

Work Experience

KeppyLab

Seattle, WA

FOUNDER

March 2024 - Present

- Created WorldEnder.ai, a text-adventure game with multiple LLMs as the backend with AI generated narratives
- Developed an AI "agent hospital" for helping undiagnosed patients find answers grounded in real medical research
- Completed Research to the People and Stanford Medicine's rare disease hackathon, presenting my team's model at GitHub HQ
- Worked with distributed, international team to develop a fine tuned RAG model for Hypophosphatasia diagnosis and research

Pegasys Medical / Fortive

Seattle, WA

CONSULTANT

September 2022 - March 2024

- Created AI pipelines for object detection and image classification
- Worked with stakeholders to gather requirements and prototype new features
- Built a robust image upload queue for synchronizing image metadata and image files across separate data stores
- Planned and implemented Golang and Python microservices for auth, file uploading, image metadata, logging
- Modernized the frontend codebase into a monorepo managed by NX—pulling all of our React/React Native applications under one repository
- Created a cost savings calculator for customer discovery and wired it up to a datastore to save user input
- Built docker compose tooling for developers

Dolly

Seattle, WA

CONSULTANT

November 2021 - August 2022

- Paired with senior engineers at Dolly to triage long-standing issues with the 'black box' payout algorithm
- Trained new hires on how to navigate the Dolly microservices and how to triage bugs that spanned across multiple products
- Fixed mobile application bugs related to outdated APIs and undocumented code
- Fixed many old, outdated Golang microservices
- Re-wrote all mono services as Node.js APIs

AI2 Incubator

Seattle, WA

ENTREPRENEUR IN RESIDENCE & AI BUILDER

December 2020 - August 2021

- Developed a computer vision AI model to detect laughter in real time during video calls
- Designed the data architecture for the Callout.ai video chat Node.js application
- Built a robust front end application in React which allowed timeline aware annotation of meetings during video calls, as well as after them
- Integrated inference models with Google Docs API to create meeting summaries using topic modeling
- Implemented several API microservices in Node.js to handle tasks like transcription and live chat over Websockets during meetings
- Utilized Pulumi to write infrastructure as code and deploy our services to AWS EC2 clusters for development, testing, and production

Moz Inc.

Seattle, WA

SR. SOFTWARE ENGINEER (LOCAL SEARCH TEAM)

May 2018 - December 2020

- Built a service in Golang to track 3rd party customer data errors—utilized the JIRA API to create and update tickets automatically as issues were solved by the customer success team
- Built a zip code specific local search optimization tool on top of Postgres with GIS plugins (React, Golang, postGIS all done together)
- Was responsible for the local search organization as a core member delivering and supporting features directly
- Participated in a company wide initiative to rewrite all microservices in Terraform and deploy them to AWS EC2 containers

RumbleMonkey

Seattle, WA

FOUNDING SOFTWARE ENGINEER

January 2017 - May 2018

- Created a FFA multiplayer game in Unity3D, working directly with a game designer. Required hand coding all player interactions, game rules, and effects
- Developed a blockchain implementation which tracked user actions and tokens in the system, allowing secure real-money transactions between users
- Went to PAX East, discovered and developed customers
- Transformed an existing desktop prototype written in Qt into a modern Electron application with an auditable, secure log of events that could be verified by the server in real time as interactions between two separate clients occurred
- Built a Golang service & API facilitating a real time betting system for the trading card game Hearthstone
- Interacted directly with customers to discover pain points in the software and solve them quickly and efficiently

Godaddy

Kirkland, WA

SOFTWARE ENGINEER

February 2015 - December 2016

- Built Ember.js applications that replaced legacy account management UIs with mobile first SPAs
- Designed and built Node.js proxy APIs that transformed legacy DB data from MySQL and made it available for consumption by product teams
- Contributed to and maintained an internal library of React components which were used across the company to implement user facing UIs
- Mentored junior developers and interns; acted as a code review gatekeeper & SCRUM master

Smashing Boxes

Durham, NC / Remote

SOFTWARE ENGINEER

July 2014 - January 2015

- Implemented Ruby APIs exposing ImageMagick functionality and job processing to front end applications
- Wrote AngularJS frontend applications for the firm's top clients
- Integrated with 3rd party payment gateways using JavaScript SDKs
- Built internal tooling for server orchestration and deployment in Ruby
- Consolidated Chef cookbooks and developed a CLI for deploying new services and running cookbooks on multiple servers at once

PugetWorks / Blue Nile

Seattle, WA

SOFTWARE ENGINEER

April 2013 - June 2014

- Wrote ClojureScript in-browser text editor leveraging the Google Closure Compiler for a startup client
- Worked at Blue Nile as a Java & JavaScript consultant in an effort to rewrite the infinite scroll UI & product search endpoints
- Implemented a handlebars-like templating language in Java so that non-technical staff could update content on the Blue Nile website
- Worked on a PostgreSQL / PostGIS & Python service to store and quickly look up millions of data points over multiple shards for a startup client

Intercon Logistics Management

Seattle, WA

SOFTWARE ENGINEER

January 2012 - April 2013

- Built out Ruby APIs for all endpoints needed in the application
- Wrote and maintained drivers for a magnetic card reader in C++
- Worked closely with PugetWorks designers to build a JavaScript UI for the Ruby backend
- Built an automated hotel check-in system for corporations with frequently traveling employees

Presentation

Seattle AI Tinkerers

Seattle, WA

WORLDENDER.AI PRESENTER

June 2024

- Introduced WorldEnder.ai, an AI driven, generative text adventure game
- Attracted contributors to the project
- Showed how the research builds upon existing work in the field, and how it can be used to create new and interesting experiences
- Showed how to RAG using AI Jeezera as context

Seattle AI Tinkerers Summer Hackathon

Seattle, WA

HACKATHON PRESENTER

June 2024

- Took WorldEnder.ai to the next level and developed a full AI driven, generative text adventure game
- Attracted more contributors to the project
- Got image generation and a full gameplay loop working
- Extended the idea of query planning to be a game making mechanism

Research to the People and Stanford Medicine's Rare Disease AI Hackathon

San Francisco, CA

DISEASE LAB PRESENTER

June 2024

- Presented my team's RAG model for helping doctors and patients who deal with Hypophosphatasia
- Ported existing knowledge graph methodologies from biomedical research to the RAG model
- Created Disease Lab, a Python package for extracting prompts and generating knowledge graphs for biomedical research

Writing

KeppyLab

AI Research Lab

FOUNDER & WRITER

June 2023 - PRESENT

- Notebooks on transformers with my Socratic dialogue notes
- Notebook & blog post about fine tuning Pythia
- Notebooks, slides, and guides on LLMs & other models as a 'game engine' in WorldEnder.ai
- Used the platform to assemble a team of researchers for Research to the People and Stanford Medicine's Rare Disease AI Hackathon
- Founder musings

Program Committees

2024 **Organizer**, KeppyLab AI Hackers & Paper Readers

Seattle, WA

2024 **Group 1 Presenter**, Daily Paper Discussion - Yannic Kilcher Discord

International

Education

Northwest University

Kirkland, WA

B.S. ENGLISH

September 2007 - July 2011

- Deep study in Shakespeare, Chaucer, and other classic English literature
- Dean's list student
- Received a scholarship for BP debate and competed at the national level on the debate team
- Tutored students at the NU writing center